AI Engineering Laboratory Manual

Lab Session 4

Comprehensive AI Course Integration and Capstone Lab

Student Edition

Lab Session 4: Comprehensive AI Course Integration and Capstone Lab

Context and lab intent

This lab is intended for final-year engineering students in an AI-focused program that emphasizes applied learning, modern AI tools, project work, and professional communication.[page:1] It is the final and most exhaustive lab block, bringing together everything students learned in the earlier labs into a full capstone-style AI system that demonstrates understanding of generative AI, local agents, retrieval, memory, workflow orchestration, evaluation, and presentation.[web:58][web:70]

Lab overview

• Lab block: Sessions 10-12

• Suggested duration: 3 hours per session, 9 hours total

• Theme: Exhaustive integration of the full course

• Focus: design, build, test, and present a complete AI solution

• Tools: ChatGPT, Groq, Docker, n8n, local Llama runtime, memory and retrieval components, and optional external tool integrations

• Pedagogy: capstone build, diagnostic testing, peer review, and formal presentation

Learning objectives

By the end of this lab, students should be able to:

• Combine chat-based GenAI, API integration, local Llama workflows, memory, and retrieval in one system.

• Choose the right AI technique for each subtask.

• Design a production-style workflow with clear stages, validation, and fallback logic.

• Use n8n agent patterns with memory and tool use where appropriate.[web:70][web:64]

• Build a complete capstone project that solves a meaningful problem.

• Test the system thoroughly using normal, edge, and failure cases.

• Present the solution clearly, including tradeoffs, limitations, and future improvements.

Prerequisites

Students should already have completed Labs 1, 2, and 3 and should have:

• A working generative AI prompt workflow.

- A working local n8n + Llama environment.
- A working end-to-end integrated AI application.
- Strong familiarity with API calls, routing, and structured outputs.

Capstone project options

Students may choose one primary capstone theme:

- AI academic assistant.
- Smart student support desk.
- Interview preparation and resume coach.
- Project brainstorming and planning assistant.
- Personalized study companion with memory.
- Document Q&A assistant with retrieval.
- Workflow automation assistant for a real campus use case.

Session structure

Session	Focus	Core outcome
10	Capstone definition and system design	Students finalize the complete solution plan
11	Build, enhance, and validate	Students implement advanced features
12	Final testing, demonstration, and review	Students present and defend the solution

Session 10

Capstone definition and system design

Session objective

Students should define a complete course-level AI solution and design the full architecture before implementation.

Concept coverage

- Problem scoping for a final-year capstone
- Selecting the right AI components for each stage
- Balancing local, API-based, and retrieval-based capabilities
- Memory, tool use, and state management
- Evaluation plan and success criteria

Instructor demonstration

The instructor should show a reference architecture that includes:

- User input.
- Request normalization.
- Intent classification.
- Generation or rewrite stage.
- Agent execution.
- Retrieval or memory when needed.
- Final answer formatting.
- Logging and monitoring.

Exercise 10.1: Finalize the problem statement

Step-by-step instructions

1. Choose a capstone theme.
2. Write the problem statement.
3. Identify the target user persona.
4. List the main pain points.
5. Define the value the AI system provides.
6. Define the success criteria for the project.

Expected outcome

Students should have a clear and meaningful capstone scope.

Exercise 10.2: Design the architecture

Step-by-step instructions

1. Draw a complete block diagram.
2. Mark local versus cloud components.
3. Identify which steps are handled by ChatGPT or Groq.
4. Identify which steps are handled by n8n and local Llama.
5. Add memory or retrieval if required.
6. Add fallback paths and logging.

Expected outcome

Students should understand the full system before building it.

Exercise 10.3: Define data contracts

Step-by-step instructions

1. Decide the input format.
2. Decide the intermediate JSON schema.
3. Decide the final response format.
4. Add validation rules.
5. Create one example for each stage.

Expected outcome

Students should understand how complex AI systems depend on clean data handoffs.

Deliverable for Session 10

- Capstone problem statement.
- Architecture diagram.
- Data contract sheet.

Session 11

Build, enhance, and validate

Session objective

Students should implement advanced features that make the capstone more robust, useful, and demonstrably complete.

Concept coverage

- Multi-step orchestration
- Memory and conversation continuity
- Retrieval from documents or knowledge notes
- Tool calling and action execution
- Robust prompt design and validation loops

Instructor demonstration

The instructor should show how to extend a basic workflow with:

- A memory node or memory manager.
- A retrieval layer using document chunks or knowledge input.
- A tool step such as email drafting, file output, or data lookup.
- Validation of outputs before delivery.

Exercise 11.1: Add memory

Step-by-step instructions

1. Add conversational memory to the AI agent where appropriate.

2. Store recent context or session preferences.

3. Test whether the workflow remembers prior user instructions.

4. Compare behavior with and without memory.

Expected outcome

Students should understand when memory improves user experience.[web:70]

Exercise 11.2: Add retrieval

Step-by-step instructions

1. Choose a local document set or knowledge base.

2. Ingest the content into the retrieval flow.

3. Ask questions that can only be answered from the source.

4. Ask questions that are outside the source.

5. Ensure the system clearly distinguishes between answerable and unanswerable cases.

Expected outcome

Students should learn retrieval-augmented workflows at a practical level.[web:68]

Exercise 11.3: Add a tool action

Step-by-step instructions

1. Add one tool to the system, such as a file writer, email draft generator, calendar item creator, or spreadsheet updater.

2. Make sure the agent can choose when to use the tool.

3. Run one successful tool-driven task.

4. Confirm the final output is still human-readable.

Expected outcome

Students should understand that agents become more useful when they can act, not just answer.[web:64][web:73]

Exercise 11.4: Validation and correction loop

Step-by-step instructions

1. Force the model to return structured JSON.

2. Validate the JSON.

3. If invalid, send it back for correction.

4. Repeat until the output is acceptable.

Expected outcome

Students should understand how to build reliability into AI workflows.

Deliverable for Session 11

- Enhanced capstone workflow.
- Memory or retrieval evidence.
- Tool-action proof.
- Validation test record.

Session 12

Final testing, demonstration, and review

Session objective

Students should thoroughly test their capstone, prepare a professional presentation, and defend their design choices.

Concept coverage

- Functional testing and edge testing
- Failure analysis
- UX quality and clarity
- Presentation and explanation
- Future extensions

Instructor demonstration

The instructor should show how to create a final test matrix with:

- Normal input.
- Ambiguous input.
- Empty input.
- Incorrect format input.
- High-complexity input.

Exercise 12.1: Build the test matrix

Step-by-step instructions

1. Create at least eight test cases.
2. Include both successful and failing inputs.

3. Note the expected result for each case.
4. Run the cases through the workflow.
5. Record actual versus expected results.

Expected outcome

Students should know how to evaluate whether their system is dependable.

Exercise 12.2: Improve the system

Step-by-step instructions

1. Identify the top three weaknesses.
2. Fix one prompt issue, one routing issue, and one formatting issue.
3. Rerun the test matrix.
4. Compare the before-and-after behavior.

Expected outcome

Students should experience iterative refinement as a normal part of AI development.

Exercise 12.3: Prepare final presentation

Step-by-step instructions

1. Create a concise slide deck or demo script.
2. Explain the problem, architecture, implementation, and results.
3. Show one live demo.
4. Explain where the system uses generative AI, local agents, retrieval, memory, and tools.
5. State what would be needed to take the project to production.

Expected outcome

Students should be able to communicate the complete system professionally.

Exercise 12.4: Viva-style reflection

Step-by-step instructions

1. Each student answers short questions from the instructor.
2. Questions should cover prompts, APIs, local models, n8n logic, memory, retrieval, and evaluation.
3. Students should explain one lesson learned from each lab block.
4. Students should state one limitation and one future improvement.

Expected outcome

Students should demonstrate conceptual understanding, not just working code.

Assessment plan

Component	Weightage
Problem definition and design	15%
Implementation completeness	25%
Memory/retrieval/tool integration	20%
Testing and refinement	20%
Presentation and viva	20%

Evaluation rubric

Criterion	Excellent	Good	Needs improvement
Capstone scope	Clear, meaningful, and well-scoped	Clear but slightly broad	Unclear or too shallow
Technical integration	Integrates multiple AI techniques effectively	Integrates most required parts	Only partial integration
Reliability	Good validation and test coverage	Basic testing present	Little evidence of testing
Explanation quality	Explains architecture and tradeoffs well	Adequate explanation	Weak understanding
Professional delivery	Strong demo and documentation	Acceptable demo and notes	Incomplete presentation

Expected cumulative outcomes

By the end of Sessions 10-12, students should be able to:

- Design and build a complete AI solution using the concepts covered in the course.
- Combine generative AI, agent workflows, retrieval, memory, and tool use into one system.
- Test and refine the solution professionally.
- Present the final result with confidence and clarity.

Suggested instructor notes

- Encourage students to keep the capstone realistic and demonstrable.

- Make the final assessment about reasoning and system design, not just feature count.
- Use the viva to verify understanding of prompts, APIs, memory, retrieval, and agents.
- Ask students to state what they would do next if they had one more week.

Materials to provide students

- Capstone planning template.
- Architecture worksheet.
- Test matrix template.
- Presentation template.
- Viva preparation questions.

Submission checklist

Students must submit:

- Final architecture and problem statement.
- Workflow export and supporting files.
- Test matrix with results.
- Final presentation materials.
- Reflection note on learnings and limitations.

Expected final lab outcome

At the end of this fourth and final lab block, students should have a complete capstone-style AI application that demonstrates mastery of the full course: generative AI, local LLMs, agents, memory, retrieval, tools, testing, and presentation.